

Euclid, Book I, Proposition I.2

To Place at a Given Point a Straight Line Equal to a Given Straight Line

CGJteamLab

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This note compares Euclid's original construction for Proposition I.2 with its reconstruction over the Hilbert foundation. In the CGJteamLab development the proposition follows immediately from Book Zero #49 (Layoff), illustrating the difference between the architectural roles of segment construction in Euclid's geometry and the Hilbert axiomatic framework.

1 Statement

Given a point A and a segment BC , construct a point D such that

$$AD \cong BC.$$

2 Euclid's Classical Construction

Euclid's construction proceeds as follows.

1. Construct the equilateral triangle DAB on the segment AB .
2. Produce the side DB beyond B .
3. Produce the side DA beyond A .
4. Draw the circle centered at B with radius BC .
5. Draw the circle centered at D with radius DG .
6. The constructed point L satisfies

$$AL \cong BC.$$

Step 1. Given data

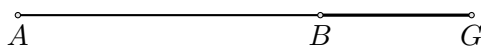
The construction starts with a given point A and a given segment BG .

$\circ$
 A

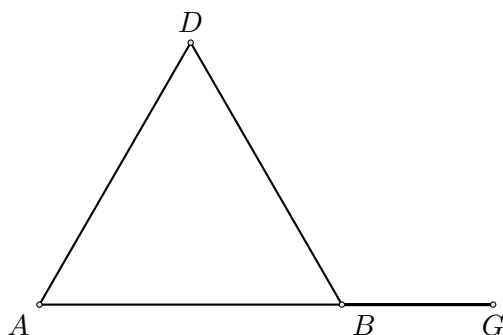
given segment
 $\circ$ ————— $\circ$
 B G

Step 2. Join AB

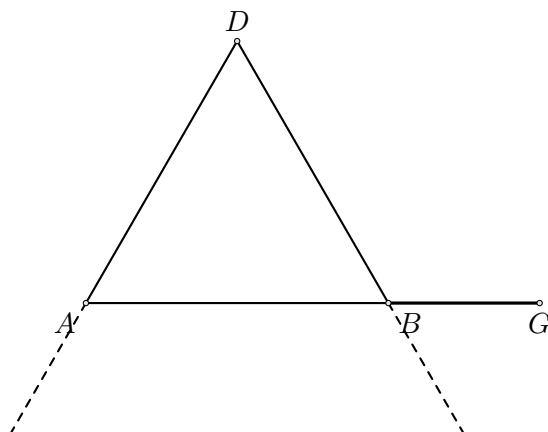
Join the given point A to the endpoint B of the given segment.

**Step 3. Construct the equilateral triangle DAB**

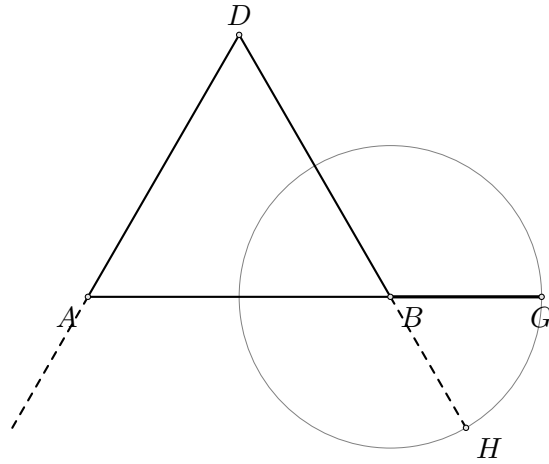
Construct the equilateral triangle DAB on the segment AB .

**Step 4. Produce DA and DB**

Produce DA beyond A and DB beyond B .

**Step 5. Draw the circle centered at B**

Draw the circle centered at B through G . Let H be the intersection of this circle with the line DB on the side beyond B .

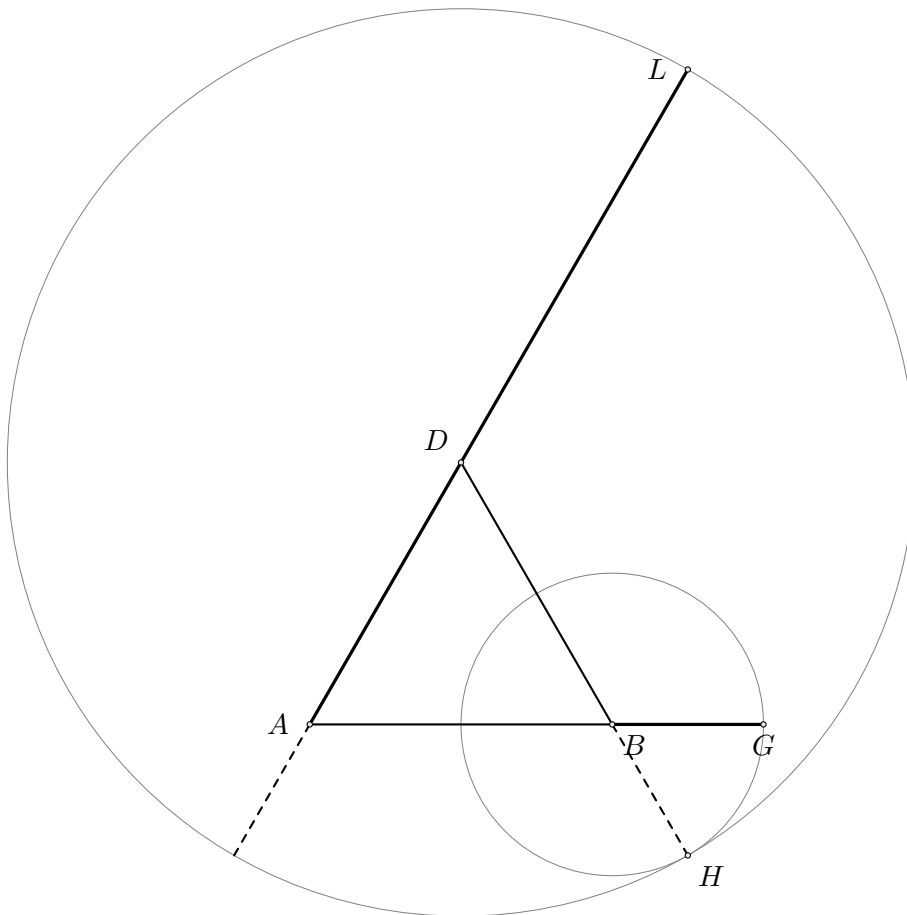


Since H lies on the circle centered at B through G ,

$$BH \cong BG.$$

Step 6. Draw the circle centered at D

Draw the circle centered at D through H . Let L be the intersection of this circle with the line DA on the side beyond A .



Since L lies on the circle centered at D through H ,

$$DL \cong DH.$$

The segment AL is therefore the required copy of the given segment.

3 Hilbert Reconstruction

The reconstruction over the Hilbert foundation is considerably shorter.

1. Choose any point $E \neq A$.
2. The ray AE is thereby determined.
3. Apply Book Zero #49 (Layoff).
4. Obtain a point D on the ray AE satisfying

$$AD \cong BC.$$

No auxiliary geometric construction is required.

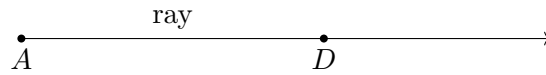


Figure 1: Lay off the segment on a chosen ray.

Book Zero #49 provides a point D on a chosen ray from A such that

$$AD \cong BC.$$

4 Formalization

The formal proof consists of a single application of Book Zero #49.

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theorem euclid_proposition_2 ...
:= bookZero_49_layoff ...
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5 Discussion

Thus Proposition I.2 illustrates a fundamental architectural difference between Euclid's geometry and the Hilbert approach.

In Euclid's *Elements*, the transport of a segment is obtained through a sequence of explicit geometric constructions beginning with Proposition I.1.

In contrast, Hilbert incorporates this operation into the congruence axioms. Book Zero encapsulates it in the Layoff Theorem, making Proposition I.2 an immediate consequence.

6 Book Zero #49: Layoff

The proof of Proposition I.2 relies on one of the elementary construction theorems established in Book Zero.

Book Zero #49 (Layoff). Let $A \neq B$ and $C \neq D$. Then there exists a point X on the ray AB such that

$$AX \cong CD.$$

In other words, every nondegenerate segment can be copied onto an arbitrarily chosen ray while preserving its length.

Input

Ray AB

A----->

Segment CD

C-----D

|

| Book Zero #49

v

Output

A-----X----->

AX congruent CD

This theorem provides the fundamental operation of transporting segments. Once it is available, Euclid's Proposition I.2 becomes an immediate consequence.

7 Relationship with Earlier Propositions

7.1 Proposition I.1

Proposition I.2 is the first substantial reuse of the equilateral triangle construction of I.1.

Euclid constructs equilateral triangles as auxiliary devices for transporting a given length to a prescribed point. The equilateral property creates rigid pairs of congruent segments from which the target segment congruence can be obtained.

Thus I.1 is not an isolated construction: its symmetry becomes working infrastructure immediately in I.2.

7.2 Transition to Proposition I.3

Proposition I.2 establishes free segment transport: given a segment and a prescribed point, a congruent segment can be constructed from that point.

Proposition I.3 adds an order constraint. When a longer segment is given, the transported length must be cut from inside it.

Hence the progression is

$$\text{I.2 : copy a length} \quad \longrightarrow \quad \text{I.3 : cut that length from a longer segment.}$$

8 Hilbert and Book Zero Components Used

8.1 Segment Layoff

The Hilbert congruence layer already contains the fundamental operation underlying I.2: a segment congruent to a given segment can be laid off from a prescribed point along a prescribed ray.

The Euclidean proposition therefore packages this stronger foundational operation into Euclid's constructional form.

8.2 Ray Construction

A segment copy requires not only a length but a direction in which the new endpoint is to be placed. The formal reconstruction represents this direction through a ray determined by distinct points and uses the Hilbert ray/order infrastructure to control the endpoint.

8.3 Segment Congruence

The target statement is expressed directly by the synthetic congruence relation

$$AB \cong CD.$$

No numerical length function is introduced.

8.4 Incidence and Nondegeneracy

The prescribed ray must be genuine, so its defining points must be distinct. The formal statement records this explicitly rather than relying on the appearance of the classical diagram.

9 Formalization Notes

Proposition I.2 reveals an early difference between Euclid’s constructional development and the Hilbert foundation.

Euclid has not yet postulated a general segment-layoff operation. Consequently he constructs the transported length indirectly using equilateral triangles and joins.

In Hilbert geometry, segment construction is already part of the congruence structure. The central mathematical content of I.2 is therefore available at a lower foundational level.

The dependency comparison is

Euclid: I.1 + auxiliary construction versus Hilbert: segment layoff.

This is analogous to what occurs much later in Proposition I.23, where Euclid derives angle transport through earlier triangle constructions while Hilbert III.4 provides angle construction directly.

Thus I.2 is the segment-level version of a recurring phenomenon in the reconstruction: a Euclidean construction theorem may become a thin interface theorem when the corresponding construction has been made foundational.

10 Dependency Summary

The classical route is

given point A and segment BC
↓
I.1 : equilateral auxiliary triangle
↓
extend and join auxiliary segments
↓
equilateral congruences + transitivity
↓
 $AD \cong BC$
↓
I.2.

The Hilbert route is conceptually shorter:

prescribed point + prescribed length + ray
↓
Hilbert segment layoff
↓
constructed endpoint
↓
required segment congruence
↓
euclid_proposition_2.

No coordinates or numerical distances are used.

11 Conclusion

Proposition I.2 establishes the transport of segment length to a prescribed point.

For Euclid this is a genuine derived construction built from the equilateral-triangle machinery of I.1. In the Hilbert reconstruction, the corresponding segment-layoff principle belongs to the foundational congruence interface.

The contrast is mathematically informative. It shows, already at the second proposition of Book I, that the historical order of Euclid's constructions need not coincide with the dependency order of a modern axiomatic reconstruction.

I.2 immediately becomes infrastructure for I.3, where free transport is combined with order to cut a prescribed length from a longer segment. It is therefore the first general segment-construction interface of the Book I development.